Bioinformatics Python Toolkit Ideas

1. Sequence Utilities (Must Have)

• gc_content.py – Calculate GC%, sliding window GC, average GC.

• reverse_complement.py – Reverse complement DNA/RNA.

• transcribe.py – DNA to RNA.

• translate.py – RNA/DNA to protein using codon table.

• orf_finder.py – Find ORFs and longest ORF.

• kmer_counter.py – Count k-mers.

• sequence_stats.py – Length, GC, AT, N count, GC skew, molecular weight.

• random_sequence.py – Generate random DNA sequences.

2. FASTA Utilities

• read_fasta.py

• write_fasta.py

• split_fasta.py

• merge_fasta.py

• rename_headers.py

• filter_fasta.py

• fasta_summary.py (count, shortest, longest, average, N50, GC%)

3. FASTQ Utilities

• fastq_stats.py

• fastq_to_fasta.py

• quality_filter.py

• read_length_distribution.py

4. Alignment Utilities

• sam_parser.py

• coverage_calculator.py

• cigar_parser.py

5. Variant Calling Utilities

• vcf_parser.py

• snp_counter.py

• variant_stats.py

• vcf_filter.py

6. Motif Analysis

- motif_search.py
- restriction_sites.py
- palindrome_finder.py

7. Comparative Sequence Analysis

- hamming_distance.py
- edit_distance.py
- pairwise_identity.py
- consensus_sequence.py

8. Population Genetics

- allele_frequency.py
- hardy_weinberg.py

9. Protein Utilities

- aa_composition.py
- protein_weight.py
- isoelectric_point.py
- hydrophobicity.py

10. Statistics

- n50.py
- gc_skew.py
- sliding_window.py

11. Visualization

- plot_gc.py
- plot_read_lengths.py
- coverage_plot.py
- variant_density.py

12. Utilities

- validate_dna.py
- clean_sequence.py
- format_sequence.py
- reverse_sequence.py
- count_bases.py

13. Advanced Scripts

- blast_result_parser.py

- gff_parser.py
- gtf_parser.py
- bed_parser.py
- bed_intersection.py
- genbank_parser.py
- gene_extractor.py
- exon_extractor.py
- promoter_extractor.py
- codon_usage.py
- multiple_sequence_alignment_stats.py
- phylogenetic_distance.py
- sequence_logo_generator.py
- genome_assembly_stats.py
- vcf_to_csv.py
- bam_flag_decoder.py
- pileup_parser.py

Recommended Learning Order

- 1. Reverse complement
- 2. GC content
- 3. FASTA reader/writer
- 4. Sequence statistics
- 5. K-mer counter
- 6. ORF finder
- 7. Translation
- 8. Motif search
- 9. FASTQ statistics
- 10. FASTQ → FASTA
- 11. VCF parser
- 12. Variant statistics
- 13. SAM parser
- 14. Coverage calculator
- 15. GFF/GTF parser
- 16. BED parser
- 17. Codon usage
- 18. N50 calculator
- 19. Consensus sequence
- 20. Genome assembly statistics